

Variational Inequality Problems

Section 7.4: Mappings Associated with Variational Inequality

Based on Pathak's Introduction to Nonlinear Analysis

Fixed Point Theory & Optimization Course

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Outline

- 1 Introduction to Variational Inequality Problems
- 2 Projection Operators and Their Properties
- 3 Normal Cones and VIP Equivalence
- 4 Fixed Point Equations and NFP Formulation
- 5 Numerical Algorithms and Applications
- 6 Existence and Uniqueness Results
- 7 Summary and Extensions

What is a Variational Inequality?

Definition (Variational Inequality Problem)

Let K be a nonempty closed convex subset of $\mathbb{R}^n$ and $F : K \rightarrow \mathbb{R}^n$ be a mapping. The **variational inequality problem** (VIP) is:

$$\text{Find } \bar{x} \in K \text{ such that } (x - \bar{x})^T F(\bar{x}) \geq 0 \text{ for all } x \in K$$

- Fundamental problem in optimization, game theory, and numerical analysis
- Unifies fixed point problems, optimization, complementarity, and equilibrium problems
- Solution interpretation: $F(\bar{x})$ is “orthogonal” to the feasible direction $x - \bar{x}$

Key Classes of VIP

- **Linear VIP:** $F(x) = Ax + b$ for matrix A and vector b
- **Nonlinear VIP:** General nonlinear mapping F
- **Monotone VIP:** F is monotone operator

Equivalent Formulations of VIP

Original Form:

$$(x - \bar{x})^T F(\bar{x}) \geq 0 \quad \forall x \in K$$

Equivalent Forms:

1. Normal Equation:

$$0 \in F(\bar{x}) + N_K(\bar{x})$$

where $N_K(\bar{x})$ is normal cone to K at $\bar{x}$.

2. Fixed Point Form:

$$\bar{x} = P_K(\bar{x} - \alpha F(\bar{x}))$$

for some $\alpha > 0$.

3. Normal-Fixed Point Equation:

$$\Psi_\alpha(x) = 0$$

where

$$\begin{aligned} \Psi_\alpha(x) = & f(P_K((1 + \alpha)x \\ & - \alpha P_K(x - \alpha f(x)))) \\ & + \alpha(P_K(x - \alpha f(x)) \\ & - P_K(x)) \end{aligned}$$

These formulations allow us to use different theoretical and computational approaches

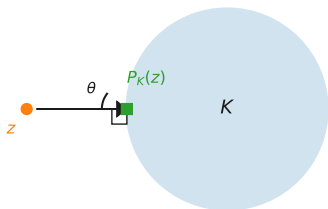
Projection Operator Definition

Definition (Projection Operator)

For a nonempty closed convex set $K \subseteq \mathbb{R}^n$ and point $z \in \mathbb{R}^n$, the **projection of z onto K** is:

$$P_K(z) = \arg \min_{y \in K} \|z - y\|^2$$

i.e., $P_K(z)$ is the unique point in K closest to z .



Key Properties of Projection

Theorem (Characterization of Projection)

Let K be a nonempty closed convex subset of $\mathbb{R}^n$ and $z \in \mathbb{R}^n$. Then $x^* = P_K(z)$ if and only if

$$(z - x^*)^T (y - x^*) \leq 0 \quad \forall y \in K$$

Important Properties

① **Non-expansiveness:**

$$\|P_K(x) - P_K(y)\| \leq \|x - y\| \quad \forall x, y$$

② **Idempotence:**

$$P_K(P_K(z)) = P_K(z) \quad \forall z$$

③ **Variational Characterization:**

$$x^* = P_K(z) \iff (x^* - z)^T (y - x^*) \leq 0 \quad \forall y \in K$$

Projection Lemma (Lemma 7.1)

Lemma (Projection Lemma)

Let K be a nonempty closed convex subset of $\mathbb{R}^n$. For any $z \in \mathbb{R}^n$, we have

$$(x - P_K(z))^T (P_K(z) - z) \geq 0 \quad \forall x \in K$$

Proof Sketch

If $z \in K$, then $P_K(z) = z$ and the inequality is trivial.

If $z \notin K$, by the projection characterization:

$$\|P_K(z) - z\|^2 = (P_K(z) - z)^T (P_K(z) - z) \leq (x - z)^T (P_K(z) - z) \quad \forall x \in K$$

The direction $P_K(z) - z$ is perpendicular to the supporting hyperplane at $P_K(z)$. Therefore, it makes an obtuse angle with any inward direction from $P_K(z)$ to $x \in K$.

Projection Operator Definition

For $z \in \mathbb{R}^n$ and closed convex set K ,

$$P_K(z) = \arg \min_{y \in K} \|z - y\|_2$$

Description

Projection Lemma (Lemma 7.1)

For nonempty closed convex $K \subseteq \mathbb{R}^n$,

$$(z - P_K(z))^T (P_K(z) - x) \geq 0$$

for all $x \in K$ and $z \in \mathbb{R}^n$

Normal Cone and VIP

Definition (Normal Cone)

Let $K \subseteq \mathbb{R}^n$ be a nonempty closed convex set and $\bar{x} \in K$. The **normal cone** to K at $\bar{x}$ is

$$N_K(\bar{x}) = \{z \in \mathbb{R}^n : (x - \bar{x})^T z \leq 0 \quad \forall x \in K\}$$

If $\bar{x} \notin K$, then $N_K(\bar{x}) = \emptyset$ by convention.

Theorem (VIP via Normal Cone)

Let $F : K \rightarrow \mathbb{R}^n$ be a mapping. Then $\bar{x}$ solves VIP if and only if

$$0 \in F(\bar{x}) + N_K(\bar{x})$$

Equivalently,

$$-F(\bar{x}) \in N_K(\bar{x})$$

$$N_K(\bar{x})$$

Example 7.1: Optimization and Normal Cones I

Example

Let ψ be a C^1 function on an open set Ω containing K . A necessary condition for $k_0 \in K$ to be a local minimizer of ψ on K is

$$0 \in d\psi(k_0) + N_K(k_0)$$

where $d\psi(k_0)$ is the gradient of ψ at k_0 .

Connection to VIP:

- 1 If k_0 is a local minimizer of ψ on K , then

$$\text{Find } k \in K \text{ such that } (k - k_0)^T \nabla \psi(k_0) \geq 0 \text{ for all } k \in K$$

This is a VIP with $F = \nabla \psi$.

- 2 The normal equation form: $0 \in \nabla \psi(k_0) + N_K(k_0)$ connects optimization to VIP.
- 3 Many optimization problems reduce to solving VIPs.

VIP as Fixed Point Problem

Theorem

Let $\bar{x}$ be a solution to VIP. Then for any $\alpha > 0$,

$$\bar{x} = P_K(\bar{x} - \alpha F(\bar{x}))$$

and conversely, if this fixed point equation holds, then $\bar{x}$ solves VIP.

Proof

Suppose $\bar{x}$ solves VIP. Let $y = \bar{x} - \alpha F(\bar{x})$ for $\alpha > 0$.

For all $x \in K$:

$$(x - \bar{x})^T F(\bar{x}) \geq 0 \quad \Rightarrow \quad (x - \bar{x})^T (\bar{x} - y) = -\alpha(x - \bar{x})^T F(\bar{x}) \leq 0$$

By projection characterization, $\bar{x} = P_K(y) = P_K(\bar{x} - \alpha F(\bar{x}))$.

Normal-Fixed Point (NFP) Equation

Definition (NFP Mapping)

The **Normal-Fixed Point (NFP)** mapping is defined as:

$$\begin{aligned}\Psi_{\alpha}(x) = & f(P_K((1 + \alpha)x - \alpha P_K(x - \alpha f(x)))) \\ & + \alpha(P_K(x - \alpha f(x)) - P_K(x)) = 0\end{aligned}$$

where $\alpha > 0$ is the stepsize parameter.

Key Results on NFP

- 1 Ψ_{α} combines projection and fixed point operators
- 2 If $\bar{x}$ solves VIP, then $\Psi_{\alpha}(\bar{x}) = 0$
- 3 If $\Psi_{\alpha}(\bar{x}) = 0$, then $\bar{x}$ solves VIP
- 4 The mapping is more amenable to iterative algorithms

Connections Between Formulations

Key Relationships:

- 1 If $\bar{x}$ solves $\text{VI}(K, F)$, then

$$\bar{x} - \frac{1}{\alpha} f(\bar{x})$$

solves a related NFP equation.

- 2 Converse: Solutions of NFP lead to solutions of VIP.
- 3 The choice of α affects convergence rate.

Theoretical Advantages:

- Use fixed point iteration theory
- Apply contraction mapping theorems
- Leverage projection properties
- Connect to monotone operator theory
- Handle constrained optimization

Computational Significance

The NFP formulation allows us to use iterative methods like:

- Picard iteration: $x_{n+1} = \Psi_{\alpha}(x_n)$
- Mann iteration with projection
- Proximal point algorithms

Projected Gradient Algorithm

Projected Gradient Method for VIP

- 1 Choose initial point $x_0 \in K$ and stepsize $\alpha > 0$
- 2 For $n = 0, 1, 2, \dots$:

$$x_{n+1} = P_K(x_n - \alpha F(x_n))$$

- 3 Repeat until convergence

Convergence Properties

- **Monotone mappings:** Guaranteed convergence under appropriate step sizes
- **Rate:** Linear (geometric) convergence for strongly monotone F
- **Practical:** Simple to implement, uses only projection and gradient
- **Stability:** Non-expansive property ensures stability

Convergence of Projection-Based Algorithm



Python Implementation: Projected Gradient Method

```
import numpy as np
from scipy.spatial.distance import cdist

def project_onto_box(x, a, b):
    """Project x onto box K = {y : a <= y <= b}"""
    return np.clip(x, a, b)

def project_gradient_vip(F, K_proj, x0,
                        alpha=0.01, max_iter=1000,
                        tol=1e-6):
    """
    Solve VIP using projected gradient method
    F: mapping
    K_proj: projection onto K
    x0: initial point
    """
    x = x0.copy()
    history = [x.copy()]

    for k in range(max_iter):
        # Gradient step
        Fx = F(x)
        x_new = K_proj(x - alpha * Fx)

        # Check convergence
        if np.linalg.norm(x_new - x) < tol:
            print(f"Converged at iteration {k}")
```

Applications of Variational Inequalities

Optimization:

- Constrained minimization
- Quadratic programming
- Convex optimization
- Nonconvex problems (locally)

Game Theory:

- Nash equilibrium
- Oligopoly models
- Traffic equilibrium
- Network games

Economics:

- Market equilibrium
- Complementarity conditions
- General equilibrium
- Spatial models

Mechanics:

- Unilateral constraints
- Friction problems
- Contact mechanics
- Elasticity

Common Problem Classes

- Monotone VI: Solution exists and is unique
- Pseudo-monotone VI: Solution exists but may not be unique

Complementarity Problems

Definition (Linear Complementarity Problem)

Given matrix $M \in \mathbb{R}^{n \times n}$ and vector $q \in \mathbb{R}^n$, find $z \in \mathbb{R}^n$ such that

$$z \geq 0, \quad Mz + q \geq 0, \quad z^T(Mz + q) = 0$$

Theorem (LCP as VIP)

The linear complementarity problem is equivalent to the VIP:

$$\text{Find } z \geq 0 \text{ such that } (w - z)^T(Mz + q) \geq 0 \text{ for all } w \geq 0$$

where $K = \mathbb{R}_+^n = \{x : x \geq 0\}$ and $F(x) = Mx + q$.

Interpretation

Complementarity: Either component is zero or the corresponding inequality is tight. Useful in optimization, economics, and control theory.

Existence Theorem for VIP

Theorem (Existence of VIP Solution)

Let $K \subseteq \mathbb{R}^n$ be a nonempty, closed, and bounded convex set, and let $F : K \rightarrow \mathbb{R}^n$ be continuous. Then the VIP has at least one solution.

Proof Idea:

- 1 Consider the fixed point equation: $\bar{x} = P_K(\bar{x} - \alpha F(\bar{x}))$
- 2 The map $P_K(\cdot - \alpha F(\cdot))$ is continuous on compact set K
- 3 By Brouwer's fixed point theorem, a fixed point exists
- 4 This fixed point solves the VIP

Uniqueness Conditions

- **Strict Monotonicity:** F strictly monotone $\Rightarrow$ unique solution
- **Strong Monotonicity:** $(x - y)^T(F(x) - F(y)) \geq \mu \|x - y\|^2$ for $\mu > 0$
- **Lipschitz Continuity:** Combined with monotonicity ensures uniqueness

Convergence Rates

Theorem (Convergence Rate for Monotone VIP)

Suppose F is monotone and Lipschitz continuous with constant L . Under appropriate stepsize $\alpha \in (0, 2/L)$, the projected gradient iteration converges.

Convergence Rates

1 Monotone (non-strongly):

$$\|x_n - \bar{x}\| = O(1/\sqrt{n})$$

(sublinear convergence)

2 Strongly monotone:

$$\|x_n - \bar{x}\| = O(e^{-\mu n})$$

(geometric/exponential convergence)

3 Pseudo-monotone: Convergence with appropriate modifications

- Better convergence rates available with advanced algorithms

Summary of Key Concepts

Core Theory:

- **VIP Definition:** Find $\bar{x} \in K$ with $(x - \bar{x})^T F(\bar{x}) \geq 0$
- **Projection:** Unique closest point in K
- **Normal Cone:** Characterizes VIP conditions
- **Equivalences:** Fixed point, normal equation, NFP
- **Existence:** Under continuity and compactness

Computational Methods:

- **Projected Gradient:** Simple, general purpose
- **Proximal Methods:** Handles non-smooth terms
- **Splitting Methods:** Decompose into subproblems
- **Acceleration:** Improve convergence rates
- **Asynchronous:** Parallel computing

Key Takeaways

VIP is a unifying framework connecting optimization, fixed points, equilibrium, and complementarity problems. The projection operator and normal cone provide geometric interpretation and practical algorithms.

Extensions and Further Topics

Theoretical Extensions:

- **Monotone Operators:** General operator-theoretic framework
- **Inclusion Problems:** $0 \in F(x) + T(x)$ with set-valued T
- **Generalized VIP:** Multiple operators and constraints
- **Stochastic VI:** Random perturbations
- **Set-Valued Mappings:** Multivalued operators

Advanced Methods:

- **Primal-Dual Algorithms:** Handle multiple operator terms
- **Coordinate Descent:** Solve component-wise
- **Mirror Descent:** Non-Euclidean geometry
- **Variationally Coherent Methods:** Exploit problem structure
- **Machine Learning:** Stochastic and online variants

Related Topics in the Course

- Chapter 5: Fixed point theorems (foundational for VIP)
- Chapter 6: Monotone operators and subdifferentials
- Chapter 7: Variational methods and optimization
- Chapter 9: Applications to differential and integral equations

References and Further Reading

Key References:

- ① Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer International Publishing.
- ② Kinderlehrer, D., & Stampacchia, G. (1980). *An Introduction to Variational Inequalities and Their Applications*. Academic Press.
- ③ Noor, M. A. (2003). Variational Inequalities and Applications. In *Encyclopedia of Mathematical Physics*.
- ④ Rockafellar, R. T., & Wets, R. J.-B. (2009). *Variational Analysis*. Springer-Verlag.

Online Resources:

- Optimization methods courses (e.g., CMU, Stanford)
- Monge-Ampère and PDE variational approaches
- Numerical methods literature

Exercise: Solving a Simple VIP

Example (Quadratic Variational Inequality)

Consider the VIP on $K = [0, 1]^n$:

Find $\bar{x} \in K$ such that $(x - \bar{x})^T A \bar{x} \geq 0$ for all $x \in K$
where A is a positive definite matrix.

Tasks:

- 1 Show that this VIP is equivalent to minimizing $\frac{1}{2} \|Ax - 0\|^2$ on K
- 2 Implement the projected gradient method
- 3 Compare convergence rates for A with different condition numbers
- 4 Verify the solution satisfies the VIP conditions